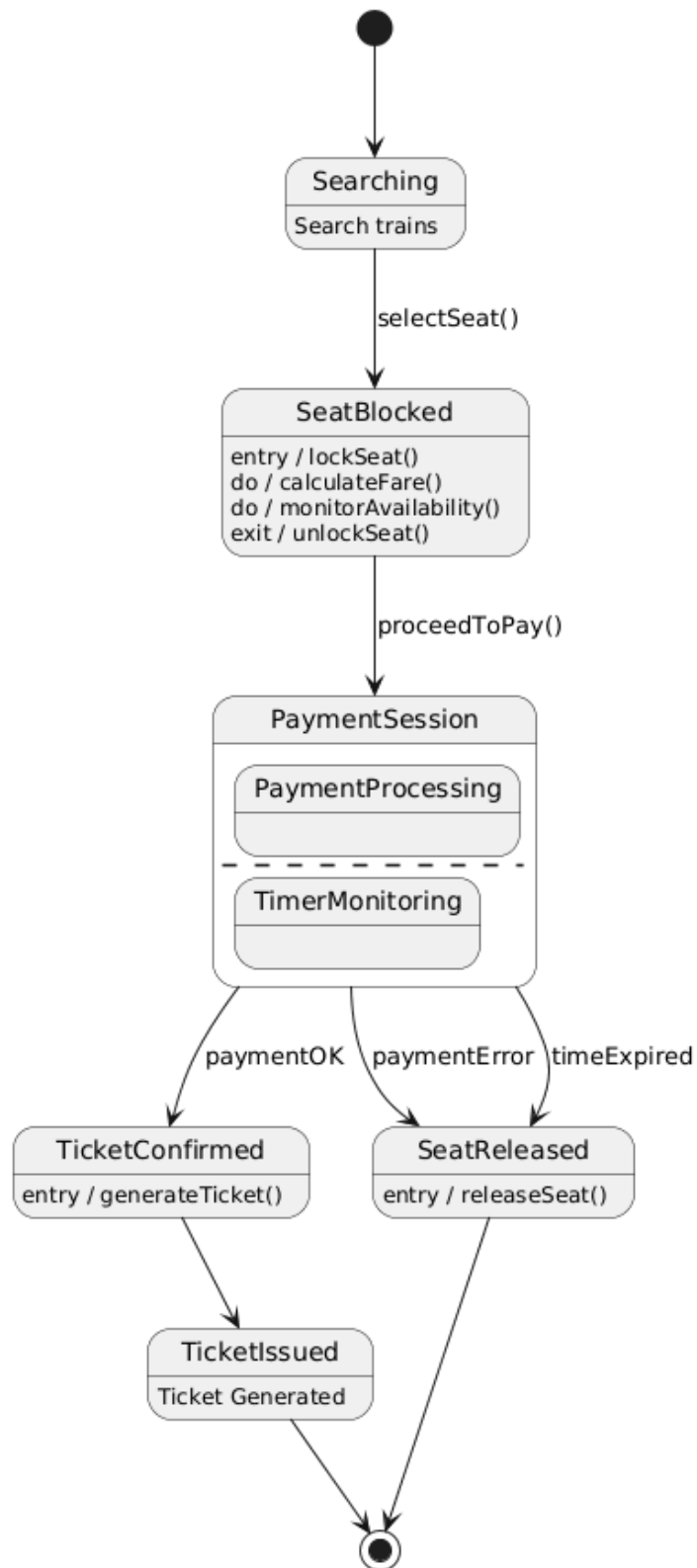


State Diagram



States Identified

1. Initial State
2. Search Train
3. Select Seat
4. Seat Blocked
5. Fare Calculation
6. Seat Availability Monitoring
7. Passenger Validation
8. Payment Processing
9. Ticket Confirmed
10. Ticket Issued
11. Seat Released
12. Final State

State	Description
Search Train	Passenger searches available trains
Select Seat	Passenger selects a seat
Seat Blocked	Selected seat is temporarily reserved
Fare Calculation	System calculates ticket fare

Seat Availability Monitoring	System continuously monitors seat status
Passenger Validation	Passenger information is verified
Payment Processing	Payment transaction is processed
Ticket Confirmed	Payment completed successfully
Ticket Issued	Ticket is generated and issued
Seat Released	Reserved seat is released
Final State	Reservation process ends

Key Transitions

- Search Train → Select Seat : Select Train
 - Select Seat → Seat Blocked : Seat Selected
 - Seat Blocked → Passenger Validation : Proceed to Payment
 - Seat Blocked → Payment Processing : Proceed to Payment
 - Payment Success → Ticket Confirmed
 - Ticket Confirmed → Ticket Issued
 - Payment Failed → Seat Released
 - Timer Expired → Seat Released
 - Ticket Issued → Final State
 - Seat Released → Final State
-

Concurrent Activities

While the system is in the **Seat Blocked** state:

Parallel Activities

- Passenger Validation
- Payment Processing

Supporting Internal Activities

- Fare Calculation
- Seat Availability Monitoring